General Comments

A broader coverage of training-free and training-based baselines

For training-free baselines, in addition to *prompt-agnostic* PruMerge and PruneSID, we also conduct experiments with one *prompt-aware* training-free baseline, SparseVLM. Here are the results on LLaVA 1.5:

method	**TextVQA**	**OCRBench**	**OCRBenchv2**	**DocVQA**	**ChartQAPro**	**Avg. Ret. (OCR)**
LLaVA-1.5-7B	58.16	18.5	23.8	25.61	12.04	100%
4x compression						
SparseVLM	55.10	15.2	21.3	17.33	11.46	85.85%
VLVLM (ours)	55.84	16.5	20.9	20.39	10.91	88.65%
8x compression						
SparseVLM	52.93	11.0	18.2	11.62	10.98	72.70%
VLVLM (ours)	55.05	15.6	20.1	17.59	10.94	84.59%

Note that our VLVLM consistently outperforms SparseVLM method, where we observe significant gains under aggressive compression budget (8x).

For training-based baselines, other than our default 1D fixed pooling, we extend our experiments on spatially local 2D pooling. Here are the results on LLaVA 1.5:

method	**TextVQA**	**OCRBench**	**OCRBenchv2**	**DocVQA**	**ChartQAPro**	**Avg. Ret. (OCR)**
LLaVA-1.5-7B	58.16	18.5	23.8	25.61	12.04	100%
4x compression						
fixed pooling	55.74	15.7	20.9	18.78	11.10	86.80%
fixed (2D)	53.37	15.7	21.9	18.60	10.67	85.97%
VLVLM (ours)	55.84	16.5	20.9	20.39	10.91	88.65%
8x compression						
fixed pooling	54.05	14.3	19.5	16.00	10.81	80.88%
fixed (2D)	51.54	13.6	20.4	16.30	9.39	77.89%
VLVLM (ours)	55.05	15.6	20.1	17.59	10.94	84.59%

The results demonstrate that 2D fixed pooling performs similarly as 1D fixed pooling, and our VLVLM maintains the best OCR-heavy task performance.

Prefill latency improvement

We measure the LLM prefill latency to demonstrate the efficiency gain beyond FLOPs, spanning a diverse collection of benchmarks (TextVQA, DocVQA for OCR-heavy tasks, MME, GQA, POPE, AND MM-Vet or general VQA tasks):

Method	**MME**	**TextVQA**	**DocVQA**	**GQA**	**POPE**	**MM-Vet**
LLaVA-1.5 7B	60.02	65.79	57.37	59.31	55.59	55.72
4x compression						
PruMerge	29.82	32.12	29.17	29.37	29.42	29.83
PruneSID	28.96	32.37	29.07	29.19	29.20	28.99
SparseVLM	42.44	46.34	41.03	41.65	42.75	41.81
Fixed pooling	29.81	33.34	29.93	30.03	30.03	29.73
VLVLM	29.62	33.34	30.00	30.23	29.78	29.82

Under 4x compression, most methods and baselines including VLVLM can reach up to **2x** prefill latency speedup with a single batch size; SparseVLM only reaches up to **1.5x** latency boost since it starts merging image tokens during the LLM layer.

To validate the latency gains with more modern VLMs, we further measure the prefill latency improvement on multiple benchmarks including OCRBench, TextVQA, and ChartQAPro for Qwen3VL 4B model. We first conduct a profiling experiment on a single-sample (batch size = 1) setting, and VLVLM achieves similar prefill speedup as other baselines such as fixed pooling and PruneSID. To maximize GPU utilization, we also consider batched (batch size = 4) setting for inference, which is not supported by PruneSID. VLVLM accelerates prefill latency similar to fixed pooling.

Method	**OCRBench**	**TextVQA**	**ChartQAPro**	**OCRBench**	**TextVQA**	**ChartQAPro**
batch size = 1				batch size = 4		
Qwen3VL-4B	92.02	69.50	131.79	406.78	299.09	830.45
4x compression						
PruneSID	57.77	53.54	63.62	--	--	--
Fixed pooling	57.76	53.87	63.49	123.85	147.11	216.54
VLVLM	58.45	54.00	63.02	124.36	143.52	215.41

Evaluation on high-resolution benchmarks

We also evaluate our method VLVLM against baselines on a more recent, real-world, high-resolution benchmark MME-RealWorld [1].

Model	monitoring	OCR	Diagram and Table	Autonomous Driving	Remote Sensing	Overall	Avg. Ret.
LLaVA	19.22	43.62	20.21	26.89	22.74	28.09	100%
4x compression							
PruMerge	19.31	42.65	20.27	26.94	22.63	27.87	99.22%
PruneSID	19.03	43.14	20.61	26.64	28.09	28.12	100.11%
fixed pooling	21.95	44.16	20.43	26.20	24.48	28.69	102.14%
VLVLM (ours)	20.26	48.24	21.79	27.49	26.86	30.77	109.54%
8x compression							
PruMerge	19.31	42.46	20.27	26.97	22.66	27.82	99.04%
PruneSID	19.31	42.89	20.49	26.64	22.74	28.03	99.79%
fixed pooling	20.95	47.06	20.89	27.35	26.57	30.20	107.51%
VLVLM (ours)	21.86	46.95	21.44	27.43	26.97	30.44	108.37%

VLVLM outperforms other baselines such as PruMerge, PruneSID, and fixed pooling, under both moderate (4x) and aggressive (8x) compression budgets, especially on tasks subsets like "OCR" and "Diagram and Table".

Coarse-grained benchmark results for Qwen3VL

We evaluate our Qwen3VL experiment on coarse-grained general VQA tasks, covering benchmarks such as MMBench, MME, GQA, and POPE. We observe that on these simple benchmarks, all methods including VLVLM and representative baselines are performing well, revealing applicability on modern state-of-the-art VLMs.

Method	MMBench	MME	GQA	POPE
Qwen3VL-4B	83.90	1710.48	62.49	89.66
SFT@66K	88.42	1729.54	62.94	90.22
Additional 4x compression				
PruneSID	85.63	1638.41	61.13	89.35
Fixed pooling@66K	85.31	1668.44	60.41	88.90
VLVLM@66K	84.97	1680.89	59.54	89.22

References

[1] MME-RealWorld: Could Your Multimodal LLM Challenge High-Resolution Real-World Scenarios that are Difficult for Humans?, ICLR 2025.